

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Introduction to Sustainable Engineering | Course Code: 25EN1120 | Credits: 3

Exam Session: Mid Semester Exam (CIA-1) March 2026 (Max Marks: 40, Duration: 15 min) | Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Compare the Linear Economy (Take-Make-Dispose) with the Circular Economy (3R: Reduce, Reuse, Recycle) model. Explain Product Stewardship and the 5 stages of Life Cycle Assessment (LCA). [8 Marks]

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Q2. State Bernoulli's equation for fluid flow. Water flows through an inclined pipe of diameter tapering from 20 cm at lower end (height 2m) to 10 cm at upper end (height 6m). If pressure at lower end is 300 kPa and flow rate is 50 L/s, determine the pressure at the upper end. [8 Marks]